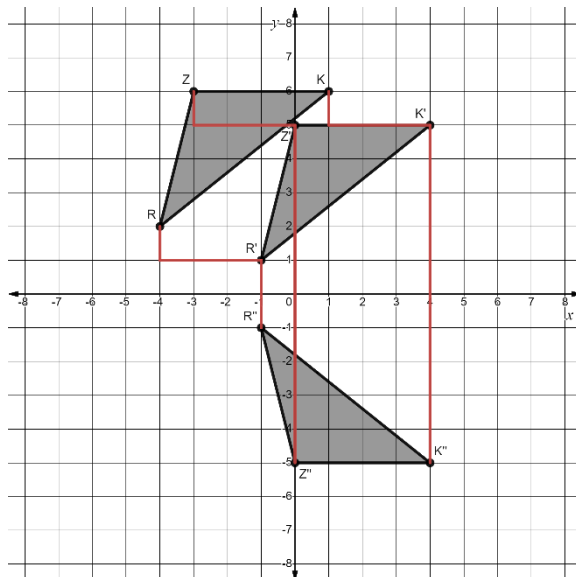


Geometry
Unit 4
Lesson 3 Homework

Name Answer Key
Date _____
Period _____

Triangles RZK , $R'Z'K'$, and $R''Z''K''$ are shown on the coordinate grid where $\triangle RZK$ is the preimage.



$$\therefore R(-4, 2) \rightarrow R'(-1, 1)$$

$$Z(-3, 6) \rightarrow Z'(0, 5)$$

$$K(1, 6) \rightarrow K'(4, 5)$$

$$x + 3, y - 1$$

$$\therefore R'(-1, 1) \rightarrow R''(-1, -1)$$

$$Z'(0, 5) \rightarrow Z''(0, -5)$$

$$K'(4, 5) \rightarrow K''(4, -5)$$

$$(x, y) \rightarrow (x, -y)$$

- Describe the feature of $\triangle R'Z'K'$ that indicates the rigid motion that was applied to $\triangle RZK$ that resulted in $\triangle R'Z'K'$. **Answers will vary.**
The location of the vertices are translated horizontally and vertically from the pre-image.
- Show the feature described by mapping the transformation of one vertex of $\triangle RZK$ to $\triangle R'Z'K'$.
- Complete the table.

Transformation	Mapping	Algebraic Description
Vertical translation down 1 unit. Horizontal translation right 3 units.	$\triangle RZK \rightarrow \triangle R'Z'K'$	$(x, y) \rightarrow (x + 3, y - 1)$

- Map the transformation of one vertex of $\triangle R'Z'K'$ to $\triangle R''Z''K''$.
- Complete the table.

Transformation	Mapping	Algebraic Description
Reflection across the x-axis.	$\triangle R'Z'K' \rightarrow \triangle R''Z''K''$	$(x, y) \rightarrow (x, -y)$

Triangles EFH and WSY are shown on the coordinate grid where $\triangle EFH$ is the preimage.

$$\therefore E(-9, -5) \rightarrow W(3, 3)$$

$$F(-6, 2) \rightarrow S(0, -4)$$

$$H(-8, -8) \rightarrow Y(2, 6)$$

$$(x, y)$$
$$(-x, -y) \rightarrow 180^\circ$$

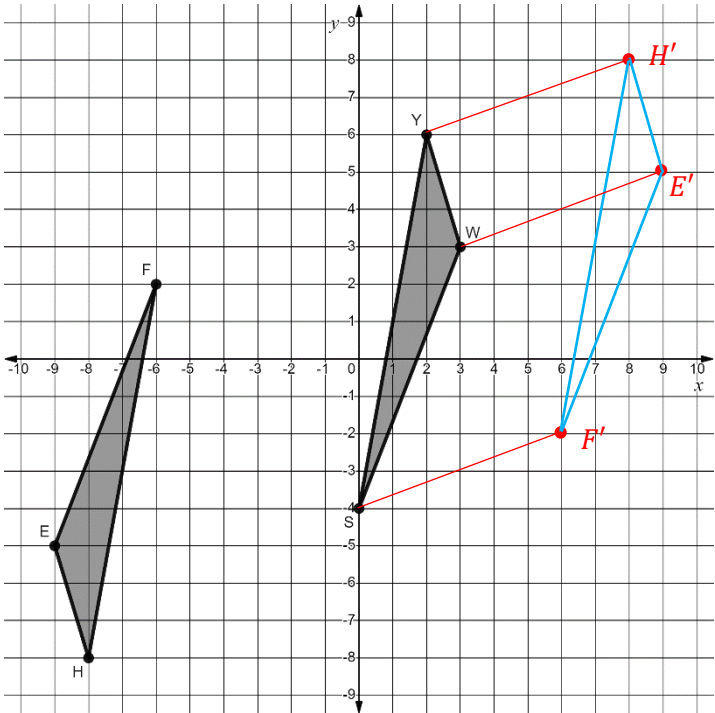
rotation $\overline{CW}$ about the origin

$$(9, 5)$$

$$(6, -2)$$

$$(8, 8)$$

$$x - 6, y - 2$$



6. Describe the feature of $\triangle WSY$ that indicates that more than one rigid motion was applied to $\triangle EFH$.
Answers will vary.
The location of the vertices are rotated and then translated horizontally and vertically from the pre-image.
7. Draw $\triangle E'F'H'$, the result of the first transformation that was applied to $\triangle EFH$.
8. Map the transformation of one vertex of $\triangle E'F'H'$ to $\triangle WSY$.
9. Complete the table.

Transformation	Mapping	Algebraic Description
180° rotation about the origin	$\triangle EFH \rightarrow \triangle E'F'H'$	$(x, y) \rightarrow (-x, -y)$
Vertical translation down 2 units. Horizontal translation left 6 units	$\triangle E'F'H' \rightarrow \triangle WSY$	$(x, y) \rightarrow (x - 6, y - 2)$

10. Explain how you know if the sequence of transformations preserves distance.
- Length of $\overline{EF} = \sqrt{58} \approx 7.6158$
Answers will vary.
- Length of $\overline{E'F'} = \sqrt{58} \approx 7.6158$
Each of the vertices on the triangles are the same
- Length of $\overline{WS} = \sqrt{58} \approx 7.6158$
distance apart after each transformation.